

Feasibility Memo Outline - Interim Submission

Week 5 - Interim Submission Dataset: Yale EMLC ED Triage

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This is an interim outline for the Feasibility Memo. The sections below highlight the planned content and supporting evidence.

1. Verdict

One sentence recommendation: proceed with caveats.

The dataset used is suitable for building a triage model, however, there are several limitations such as class imbalance, outliers and validity that needs to be respected from the beginning.

This section will be written last, after the key concerns and reasons for proceeding have been processed and reviewed.

2. Dataset Summary

- a. There are approximately 55,121 emergency department (ED) encounters.
- b. It contains about 225 input features plus an ESI triage label.
- c. There are 200 binary chief complaint indicators.

ESI Distribution

- a. Highly concentrated in mid-acuity patients (ESI Level 3).
- b. Very few extreme cases were presented

Patient Mix

- a. Includes age, gender, race ethnicity and insurance status.
- b. Reflects a US hospital population rather than a Caribbean ED.

The temperature is recorded in Fahrenheit and also later converted to Celsius during preprocessing.

3. Top 3 Concerns

Severe Class imbalance

Very few ESI level 1 and ESI Level 5 cases were presented. This will make it difficult for the model to learn rare but clinically important outcomes.

Data Quality Issues

The dataset missingness is minimal, however, some vitals include impossible or extreme values identified during the outlier screening.

Limited External Validity

The dataset comes from a US hospital, which may not reflect patient demographics or resource constraints in a Caribbean ED

Source: Data Profiling outputs

4. Top 3 Reasons to Proceed

Complete Structured Dataset

The dataset contains key fields such as demographics and ESI labels which are fully populated and provides a strong foundation for modelling without any missing data.

Clinically Consistent Signals

Vital signs and high-acuity complaints were present and it behaved as expected. This suggest that the dataset contains meaningful structure.

Leakage Being Removed

The dataset contained information that would only be known after triage. This has been identified and excluded from the dataset.

Source: Data Profiling and Exploratory Visualisation

5. Caveats

- a. Temperature consistency (Fahrenheit and Celsius) must be maintained throughout the modelling
- b. Chief-complaint indicators are highly sparse, with many features appearing rarely or not at all in the dataset.
- c. Correlations show the relationship between variables but it does not imply the cause-and-effect or a reliable predictive power.
- d. The model performance may not generalise well to a Caribbean ED due to the difference in patient population and the healthcare system context.